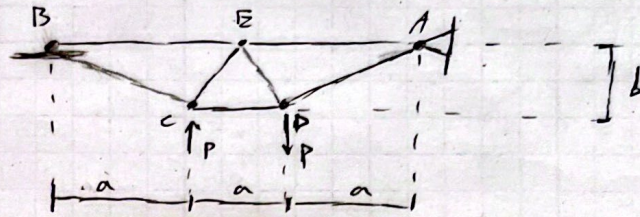


Truss Structure:



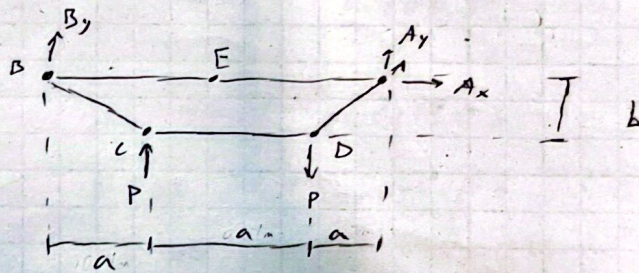
Using Constraints: $a = 0.4 \text{ m}$
 $b = 0.3 \text{ m}$
 $P = 20 \text{ kN}$

Support B is a roller support
 Support A is a pin support

External FBD:

$$\sum F_x = 0 = A_x$$

$$\boxed{A_x = 0}$$



$$\sum M_A = 0 = -B_y(3a) - P(2a) + P(a)$$

$$B_y = -\frac{2P}{3} + \frac{P}{3}$$

$$B_y = -\frac{2(20 \text{ kN})}{3} + \frac{(20 \text{ kN})}{3}$$

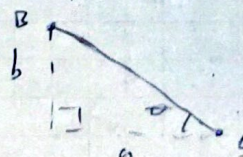
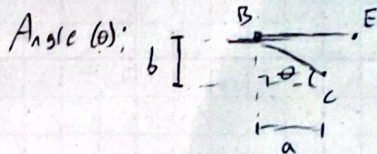
$$\boxed{B_y = -6.67 \text{ kN, downward}}$$

$$\sum M_B = 0 = A_y(3a) - P(2a) + P(a)$$

$$A_y = -\frac{P}{3} + \frac{P(2)}{3}$$

$$A_y = -\frac{(20 \text{ kN})}{3} + \frac{(20 \text{ kN})(2)}{3}$$

$$\boxed{A_y = 6.67 \text{ kN, upward}}$$

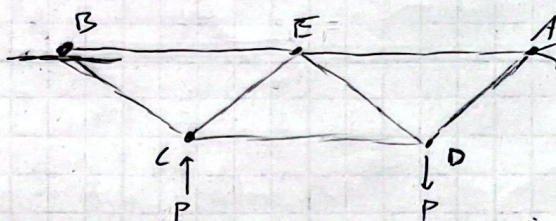


$$\theta = \tan^{-1}\left(\frac{b}{a}\right)$$

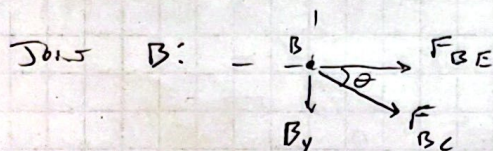
$$\theta = \tan^{-1}\left(\frac{0.3 \text{ m}}{0.4 \text{ m}}\right)$$

$$\boxed{\theta = 36.87^\circ}$$

Joint Analysis:



$$\theta = 36.87$$



$$\sum F_y = 0 = -B_y - F_{BC} \sin(\theta)$$

$$F_{BC} = \frac{-B_y}{\sin(\theta)}$$

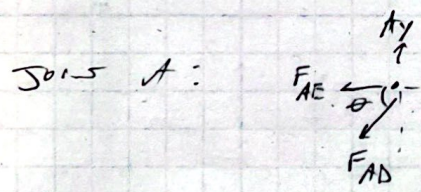
$$F_{BC} = \frac{-6.67 \text{ kN}}{\sin(36.87^\circ)} = \boxed{11.12 \text{ kN Compression}}$$

$$\sum F_x = 0 = F_{BE} + F_{BC} \cos(\theta)$$

$$F_{BE} = -F_{BC} \cos(\theta)$$

$$F_{BE} = -((11.12 \text{ kN}) \cos(36.87^\circ))$$

$$\boxed{F_{BE} = 8.89 \text{ kN Tension}}$$



$$\sum F_y = 0 = A_y - F_{AD} \sin(\theta)$$

$$F_{AD} = \frac{A_y}{\sin(\theta)}$$

$$F_{AD} = \frac{(6.67 \text{ kN})}{\sin(36.87^\circ)}$$

$$\boxed{F_{AD} = 11.12 \text{ kN Tension}}$$

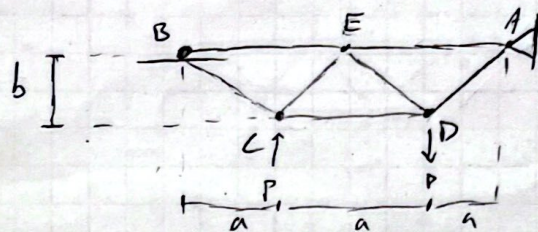
$$\sum F_x = 0 = -F_{AD} \cos(\theta) - F_{AE}$$

$$F_{AE} = -F_{AD} \cos(\theta)$$

$$F_{AE} = -(11.12 \text{ kN} \cos(36.87^\circ))$$

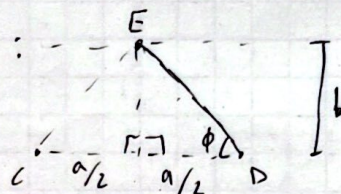
$$\boxed{F_{AE} = 8.89 \text{ kN compression}}$$

Joint Analysis (Cont.)



$$\theta = 36.87$$

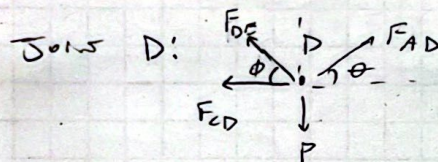
Angle (ϕ):



$$\phi = \tan^{-1} \left(\frac{b}{a/2} \right)$$

$$\phi = \tan^{-1} \left(\frac{(0.3m)}{(\frac{0.4m}{2})} \right)$$

$$\boxed{\phi = 56.31^\circ}$$



$$\sum F_y = 0 = -P + F_{AD} \sin(\theta) + F_{DE} \sin(\phi)$$

$$F_{DE} = \left[\frac{P - F_{AD} \sin(\theta)}{\sin(\phi)} \right]$$

$$F_{DE} = \left[\frac{(20kN) - (11.12kN \sin(36.87^\circ))}{\sin(56.31^\circ)} \right]$$

$$\boxed{F_{DE} = 16.02 \text{ kN Tension}}$$

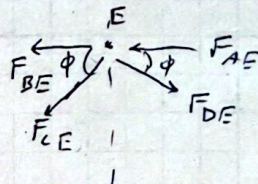
$$\sum F_x = 0 = F_{AD} \cos(\theta) - F_{DE} \cos(\phi) - F_{CD}$$

$$F_{CD} = F_{AD} \cos(\theta) - F_{DE} \cos(\phi)$$

$$F_{CD} = 11.12 \text{ kN} \cos(36.87^\circ) - 16.02 \text{ kN} \cos(56.31^\circ)$$

$$\boxed{F_{CD} \approx 0 \text{ kN}} \checkmark$$

Joint E:



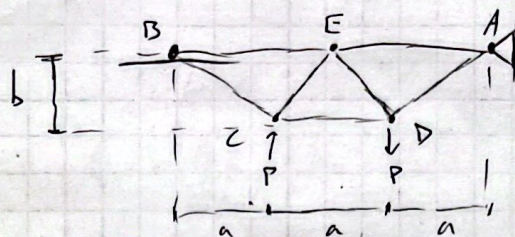
$$\sum F_x = 0 = F_{DE} \cos(\phi) - F_{AE} - F_{BE} - F_{CE} \cos(\phi)$$

$$F_{CE} = \left[\frac{F_{DE} \cos(\phi) - F_{AE} - F_{BE}}{\cos(\phi)} \right]$$

$$F_{CE} = \left[\frac{16.02 \text{ kN} \cos(56.31^\circ) - 8.89 \text{ kN} - 8.89 \text{ kN}}{\cos(56.31^\circ)} \right]$$

$$\boxed{F_{CE} = 16.07 \text{ kN compression}}$$

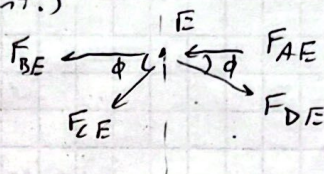
Joint Analysis (cont.)



$$\theta = 36.87^\circ$$

$$\phi = 56.31^\circ$$

Joint E (cont.)

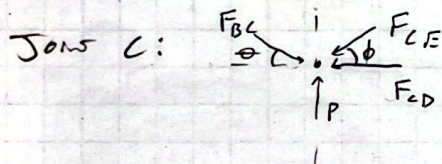


$$\sum F_y = 0 = -F_{DE} \sin(\phi) - F_{CE} \sin(\phi)$$

$$0 \approx -(16.02 \text{ kN} \sin(56.31^\circ))$$

$$-(16.07 \text{ kN} \sin(56.31^\circ))$$

$$\boxed{0 \approx 0} \checkmark$$



$$\sum F_y = 0 = P - F_{BC} \sin(\theta) - F_{CE} \sin(\phi)$$

$$0 = (20 \text{ kN}) - 11.12 \text{ kN} \sin(36.87^\circ) - (16.07 \text{ kN} \sin(56.31^\circ))$$

$$\boxed{0 \approx 0} \checkmark$$

$$\sum F_x = 0 = F_{BC} \cos(\theta) - F_{CE} \cos(\phi) - F_{CD}$$

$$0 = 11.12 \text{ kN} \cos(36.87^\circ) - 16.07 \text{ kN} \cos(56.31^\circ) - 0$$

$$\boxed{0 \approx 0} \checkmark$$

Force Table

MEMBER	Symbolic	Value & Force Type
BC	$F_{BC} = -\frac{B_y}{\sin(\theta)}$	11.12 kN, Compression
BE	$F_{BE} = -F_{BC} \cos(\theta)$	8.896 kN, Tension
AD	$F_{AD} = \frac{A_y}{\sin(\theta)}$	11.12 kN, Tension
AE	$F_{AE} = -F_{AD} \cos(\theta)$	8.896 kN, Compression
DE	$F_{DE} = \left[\frac{P - F_{AD} \sin(\theta)}{\sin(\phi)} \right]$	16.02 kN, Tension
CD	$F_{CD} = F_{AD} \cos(\theta) - F_{DE} \cos(\phi)$	0 kN
CE	$F_{CE} = \left[\frac{F_{DE} \cos(\phi) - F_{AE} - F_{BE}}{\cos(\phi)} \right]$	16.07 kN, Compression

External Forces :

B_y	$-\frac{2P}{3} + \frac{P}{3}$	6.67 kN, downward
A_y	$-\frac{P}{3} + \frac{P(2)}{3}$	6.67 kN, upward
A_x	$\{F_x = 0\}$	0 kN

Angles :

θ	$\angle BC \text{ and } \angle AD$	$\theta = \tan^{-1}\left(\frac{b}{a}\right) = 36.87^\circ$ From horizontal
ϕ	$\angle ED \text{ and } \angle EC$	$\phi = \tan^{-1}\left(\frac{b}{(a/2)}\right) = 56.31^\circ$ From horizontal

Stress Calculations: MEMBERS USING Solidworks Materials:

A500 [~] ASTM A36 steel
structural steel

Knowns:

- $FOS_{member} = 3.5$
- $F_{internal\ largest} = 16.07\text{ kN} = F_{CE}$
- Material Density of ASTM A36 steel
Density = $7850 \frac{\text{kg}}{\text{m}^3}$
- σ_{yield} of ASTM A36 steel
 $\sigma_{yield} = 250\text{ MPa} \rightarrow 36257.9\text{ psi}$
- $A = s^2$ (Square cross-section)

Unknowns:

- σ_{allow} of ASTM A36 steel
- A_{min} due to σ_{allow}
- I_{min} due to A_{min}
- Total Volume (V) of Truss
- Weight of Truss

Solving For A_{min} :

$$\sigma_{allow} = \frac{F}{A_{min\ member}} \quad (Eq 1)$$

$$\sigma_{allow} = \frac{\sigma_{yield}}{FOS} = \frac{F}{A_{min}} \quad (Eq 2)$$

$$A_{min\ member} = F \left(\frac{FOS}{\sigma_{yield}} \right)$$

$$A_{min\ member} = (16.07 \cdot 10^3\text{ N}) \left(\frac{3.5}{250\text{ MPa}} \right)$$

Square cross-section:

$$A_{min\ member} = s^2$$

$$s = \sqrt{224.98\text{ mm}^2} = 15\text{ mm}$$

$$\left| \begin{array}{l} A_{min\ member} = 224.98\text{ mm}^2 \\ = 0.349\text{ m}^2 \end{array} \right|$$

Solving For σ_{allow} : $\sigma_{allow} = \frac{F}{A_{min\ member}}$

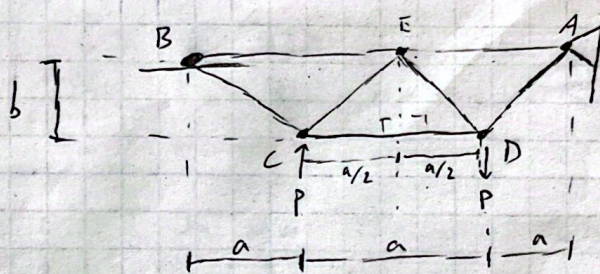
$$\sigma_{allow} = \frac{(16.07 \cdot 10^3\text{ N})}{(224.98\text{ mm}^2)} = 71.43\text{ MPa}$$

↓
10359.8 psi

Stress Calculations (Cont.): MEMBERS

- To ensure stability and realism: Each member will need to have some distance from its respective edge, where the pin is located. This length will be known as (L_0)
- To ensure proper CAD Model dimensions: Each member needs to have a calculated center-to-center of pin slots length. This length will be (L)

Solving For center-to-center lengths (L): MEMBERS OF TRUSS



Solving Top portion length: (BE & EA)

$$L_{BE} = L_{EA} = a + \frac{a}{2} = 0.4m + \frac{0.4m}{2} = 0.6m$$

$$\boxed{\begin{matrix} L_{BE} = 0.6m \\ L_{EA} = 0.6m \end{matrix}}$$

Solving For Bottom portion length: (CD)

$$L_{CD} = a = 0.4m$$

$$\boxed{L_{CD} = 0.4m}$$

Solving For Diagonal portion length (outer): (BC & AD)

$$L_{BC} = L_{AD} = \sqrt{(b)^2 + (a)^2}$$

$$L_{BC} = L_{AD} = \sqrt{(0.3m)^2 + (0.4m)^2}$$

$$L_{BC} = L_{AD} = 0.5m$$

$$\boxed{\begin{matrix} L_{BC} = 0.5m \\ L_{AD} = 0.5m \end{matrix}}$$

Solving For Diagonal portion length (inner): (CE & DE)

$$L_{CE} = L_{DE} = \sqrt{(b)^2 + (\frac{a}{2})^2}$$

$$L_{CE} = L_{DE} = \sqrt{(0.3)^2 + (\frac{0.4m}{2})^2} = 0.361m$$

$$\boxed{\begin{matrix} L_{CE} = 0.361m \\ L_{DE} = 0.361m \end{matrix}}$$

Stress Calculations (cont.): MEMBERS

Solving For Total Center-to-Center Length (L):

$$Total(L) = L_{BE} + L_{AE} + L_{BC} + L_{AD} + L_{CD} + L_{CE} + L_{DE}$$

$$L = (0.6m) + (0.6m) + (0.5m) + (0.5m) + (0.4m) + (0.361m) + (0.361m)$$

$$\boxed{L = 3.322m} \leftarrow \text{Center-to-Center Length Total}$$

• Now This Center-to-Center length (L) is Found
This length (L) along side Edge-to-Edge length (L₀)
will be used For The CAD Model.

• In order To Solve For (L₀) (Edge-to-Edge)
Need To calculate pin diameter First

• After pin diameter is calculated, (L₀) can be calculated.

• Since (L₀) covers Edge-to-Edge length of Each Member,
This length is Needed For Weight of Truss.

• To get Weight of Truss, we need Volume,
• To get volume, we need (L₀) $\rightarrow (V = L_0 \cdot A_{pin})$
• To get (L₀), we need diameter of pin First

Stress Calculations (cont.): PINS

Unknowns: For pin

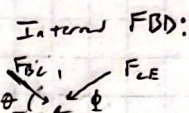
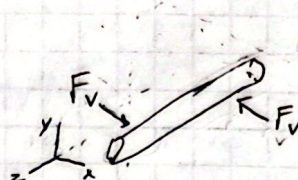
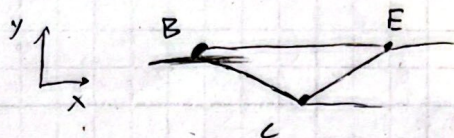
- T_{allow}
- $A_{pin} (min)$
- $W (Weight)$ of pins combined
- L_0 (Edge-to-Edge)
- Volume (V)

Knowns: (For pin)

- $FOS_{pin} = 4$
- $T_{yield} = 170 \text{ ksi}$
- Density (ρ_{pin}) = $0.278 \frac{lb}{in^3}$
- No Buckling stress within any part in compression
- $F_{internal\ force} = 16.67 \text{ kN} = F_{CE}$
- Material of pin:
Hardened Tool Steel
- Pins Experience Compression
- Single Shear connection

Stress calculations (cont.) : PINS

FBD of pin with $F_{CE} = F_{\text{internal force}}$



At joint C, F_{CE} is under compression

Force vector F_v is both inward

$$F = F_{\text{internal max}} = 16.07 \text{ kN} \rightarrow 3.613 \text{ kips}$$

$$\rightarrow (3.613 \cdot 10^3 \text{ lb})$$

Solving For Cross-Sectional Area of Pins:

$$\tau_{\text{allow}} = \frac{\tau_{\text{yield}}}{FOS_{\text{pin}}} \quad (\text{Eq 1})$$

$$\tau_{\text{allow}} = \frac{F}{A_{\text{min pin}}} = \frac{\tau_{\text{yield}}}{FOS_{\text{pin}}} \quad (\text{Eq 2})$$

$$A_{\text{min pin}} = F \left(\frac{FOS_{\text{pin}}}{\tau_{\text{yield}}} \right)$$

$$A_{\text{min pin}} = (3.61 \cdot 10^3 \text{ lb}) \left(\frac{4}{(170 \cdot 10^3) \text{ psi}} \right)$$

$$|A_{\text{min pin}} = 0.085 \text{ in}^2| \rightarrow \boxed{54.89 \text{ mm}^2}$$

$$A_{\text{min pin}} = \frac{\pi}{4} d_{\text{pin}}^2$$

$$d_{\text{pin}} = \sqrt{A_{\text{min pin}} \left(\frac{4}{\pi} \right)}$$

$$d_{\text{pin}} = \sqrt{(0.085 \text{ in}^2) \left(\frac{4}{\pi} \right)}$$

$$\boxed{d_{\text{pin}} = 0.33 \text{ in}}$$

$$\rightarrow \boxed{8.38 \text{ mm}}$$

Solving For Length

of pins:

$$\sigma_{\text{bearing}} = \frac{F}{A_b} = \frac{F}{d \cdot L} \quad (\text{Eq 1})$$

$$\sigma_{\text{bearing pin}} = \frac{F}{A_b} = \frac{F}{d_{\text{pin}} \cdot L_{\text{pin}}} \quad (\text{Eq 3})$$

$$\sigma_{\text{bearing pin}} = \frac{\sigma_{\text{yield (member)}}}{FOS_{\text{(member)}}} \quad (\text{Eq 2})$$

$$L_{\text{min}} = \frac{F (FOS_{\text{member}})}{\sigma_{\text{yield member}} \cdot d_{\text{pin}}} = \frac{(3.61 \cdot 10^3 \text{ psi})(3.5)}{(36257.4 \text{ psi})(0.33 \text{ in})}$$

$$|L_{\text{min}} = 1.06 \text{ in}| \rightarrow \boxed{26.924 \text{ mm}}$$

Now This d_{pm} is Found
Pin Slot offset: We can Find (L_o)

Length (L) is calculated From Center-to-Center
of pins From Each Member

Length (L_o) is length of Each Member
From Edge - to Edge

Structural Engineering Standard:

$$\text{Pin Slot offset} = 2 \cdot d_{pm}$$

$$\text{Pin Slot offset} = 2 \cdot (8.38 \text{ mm})$$

$$\text{Pin Slot offset} = \boxed{16.76 \text{ mm}}$$

From Each
Respective Edge
of Each Member

Length (L_o) of Each Member: (Member has 2 Edges)

Top: $L_{BE_o} = L_{BE} + (2 \cdot \text{Pin Slot offset})$

$$L_{BE_o} = L_{EA_o} = 460 \text{ mm} + (2 \cdot 16.7 \text{ mm})$$

$$\boxed{\begin{array}{l} L_{BE_o} = 633.85 \text{ mm} \\ L_{EA_o} = 633.85 \text{ mm} \end{array}}$$

Bottom: $L_{CD_o} = L_{CD} + (2 \cdot \text{Pin Slot offset})$

$$L_{CD_o} = 400 \text{ mm} + (2 \cdot 16.7 \text{ mm})$$

$$\boxed{L_{CD_o} = 433.85 \text{ mm}}$$

Diagonal Portion (outer):

$$L_{BC_o} = L_{BC} + (2 \cdot \text{Pin Slot offset})$$

$$L_{BC_o} = L_{AD_o} = 500 \text{ mm} + (2 \cdot 16.7 \text{ mm})$$

$$\boxed{\begin{array}{l} L_{BC_o} = 533.85 \text{ mm} \\ L_{AD_o} = 533.85 \text{ mm} \end{array}}$$

Diagonal Portion (inner):

$$L_{CE_o} = L_{CE} + (2 \cdot \text{Pin Slot offset})$$

$$L_{CE_o} = L_{DF_o} = 361 \text{ mm} + (2 \cdot 16.7 \text{ mm})$$

$$\boxed{\begin{array}{l} L_{CE_o} = 394.52 \text{ mm} \\ L_{DF_o} = 394.52 \text{ mm} \end{array}}$$

STRESS CALCULATIONS

MEMBERS :

	Symbolic	Values
Area member	$A = F \left(\frac{F_{0.3}}{\sigma_{yield}} \right)$	$224.98 \text{ mm}^2 \rightarrow 0.347 \text{ in}^2$
Side (s)	$A = s^2$	$15 \text{ mm} \rightarrow 0.6 \text{ in}$
Stress member	$\sigma = \frac{F}{A}$	$71.43 \text{ MPa} \rightarrow 10357.8 \text{ psi}$

MEMBER LENGTHS: Length (L)

	Symbolic	Values
L_{BE}	$a + \frac{a}{2}$	$0.6 \text{ m} \rightarrow 600 \text{ mm}$
L_{EA}	$a + \frac{a}{2}$	$0.6 \text{ m} \rightarrow 600 \text{ mm}$
L_{CD}	a	$0.4 \text{ m} \rightarrow 400 \text{ mm}$
L_{BC}	$\sqrt{(b)^2 + (a)^2}$	$0.5 \text{ m} \rightarrow 500 \text{ mm}$
L_{AD}	$\sqrt{(b)^2 + (a)^2}$	$0.5 \text{ m} \rightarrow 500 \text{ mm}$
L_{CE}	$\sqrt{(b)^2 + \left(\frac{a}{2}\right)^2}$	$0.361 \text{ m} \rightarrow 361 \text{ mm}$
L_{DE}	$\sqrt{(b)^2 + \left(\frac{a}{2}\right)^2}$	$0.361 \text{ m} \rightarrow 361 \text{ mm}$
L_{TOTAL}	$\sum L$	$3.322 \text{ m} \rightarrow 3322 \text{ mm}$

PINS :

	Symbolic	Values
Area pin	$F \left(\frac{F_{0.3 \text{ pin}}}{\sigma_{yield}} \right)$	$0.085 \text{ m}^2 \rightarrow 54.89 \text{ mm}^2$
d_{pin}	$\sqrt{A_{pin}} \left(\frac{4}{\pi} \right)$	$0.33 \text{ m} \rightarrow 8.38 \text{ mm}$
$L_{pin \text{ pin}}$	$\frac{F(F_{0.3 \text{ member}})}{\sigma_{yield \text{ member}} \cdot d_{pin}}$	$1.06 \text{ m} \rightarrow 26.929 \text{ mm}$
pin slot offset	$2 \cdot d_{pin}$	16.76 mm

Stress Calculations (Cont.): MEMBERS

Solving For Volume (V) using Edge-to-Edge length (L_o):

$$\text{Total length } (L_{o \text{ TOTAL}}) = L_{DE_o} + L_{EA_o} + L_{CD_o} + L_{BC_o} + L_{AB_o} + L_{CE_o} + L_{DE_o}$$

$$L_{o \text{ TOTAL}} = (633.85 \text{ mm}) + (633.85 \text{ mm}) + (433.85 \text{ mm}) + (533.85 \text{ mm}) + (533.85 \text{ mm}) + (394.52 \text{ mm}) + (394.52 \text{ mm})$$

$$L_{o \text{ TOTAL}} = 3558.29 \text{ mm} \rightarrow 3.56 \text{ m}$$

Now Solve Volume (V) = $L_o \cdot A$

$$A = 224.98 \text{ mm}^2$$

$$V = L_o \cdot A$$

$$V = (3558.29 \text{ mm}) \cdot (224.98 \text{ mm}^2)$$

$$V = (3.56 \text{ m}) \cdot (224.10^{-4} \text{ m}^2)$$

$$V = 7.97 \cdot 10^{-7} \text{ m}^3$$

Using Volume $\rightarrow$ Mass = $V(\text{density})$

$$\text{Mass} = (7.97 \cdot 10^{-7} \text{ m}^3) (7850 \frac{\text{kg}}{\text{m}^3})$$

$$\text{Mass} = 6.26 \text{ kg}$$

$$6.26 \text{ kg} \rightarrow 13.8516 \text{ (Weight)}$$

$$\text{Weight} = 13.8516$$

Now That We Took Into account The pin Slot offset distance
This more accurately reflects Both CAD Model
and Realism.

Now We can move on To Find Weight of pins:

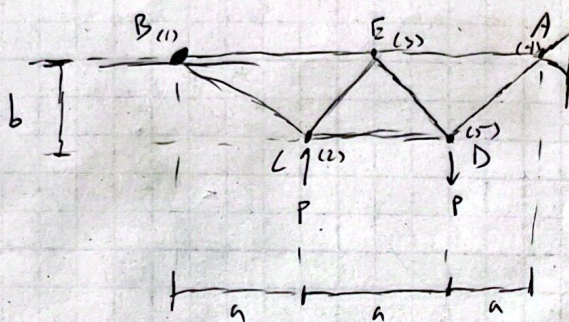
Solving For Combined weight of pins (cont.):

$$W_{pin} = n_{pins} [A_{pin} \cdot L_{pin} \cdot \rho_{pin}]$$

$$W_{pin} = 5 \left[0.085 \text{ in}^2 \cdot 1.06 \text{ in} \cdot 0.273 \frac{\text{lb}}{\text{in}^3} \right]$$

$$W_{pin} = 0.123 \text{ lb for 5 pins (theoretical)}$$

$$M_{mass \text{ pin}} = 0.056 \text{ kg}$$



Total weight of Truss: Theoretical

$$W_{TOTAL} = W_{pin} + W_{member}$$

$$W_{TOTAL} = (0.123 \text{ lb}) + (13.85 \text{ lb})$$

$$W_{TOTAL} = 13.973 \text{ lb}$$

Total weight of Truss: CAD Analysis

$$W_{member} : 5.883 \text{ kg} \rightarrow 12.97 \text{ lb}$$

$$W_{pin} : (1.143 \cdot 10^{-2} \text{ kg}) \rightarrow [0.02519 \text{ lb}] \cdot 5 = 0.126 \text{ lb for 5 pins}$$

$$W_{TOTAL} = W_{member} + W_{pin}$$

$$W_{TOTAL} = 12.97 \text{ lb} + 0.126 \text{ lb}$$

$$W_{TOTAL} = 13.096 \text{ lb}$$

Percentage Error: From Assembly File

$$(\text{Weight}): \left| \frac{13.1 - 13.973}{13.1} \right| = 6.7\%$$

$$(\text{mass}): \left| \frac{5.94 - 6.316}{5.941} \right| = 6.31\%$$

STRESS CALCULATIONS CONT.

MEMBERS LENGTH (L_o):

	<u>Symbolic</u>	<u>Value</u>
L_{BEo}	$L_{member} + (2 \cdot \text{pin SNT offset})$	633.85mm
L_{EAo}		633.85mm
L_{CDo}		433.85mm
L_{BCo}		533.85mm
L_{ADo}		533.85mm
L_{CEo}		394.52mm
L_{DEo}		394.52mm
L_{TTL}	ΣL_o	3558.27mm

MEMBERS OTHER:

	<u>Symbolic</u>	<u>Values</u>
Volume (V)	$L_o \cdot A$	$7.97 \cdot 10^{-4} m^3$
Mass	$V (\text{density})$	$6.26kg \rightarrow 13.85lb$ (Wacker)

PINS OTHER:

	<u>Symbolic</u>	<u>Value</u>
W_{pin}	$n_{pins} [A_{pin} \cdot L_{pin} \cdot \rho_{pin}]$	0.123lb Per 5 pins